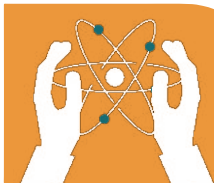


UNIT 9

FORCE AND ITS EFFECTS

Learning objectives:

- Understand force and its impact on motion, shape, and size.
- Explain Hooke's Law and its relation to force, extension, and spring constant.
- Analyze pressure and its role in hydraulic systems and Pascal's Law.



FORCE AND ITS EFFECTS

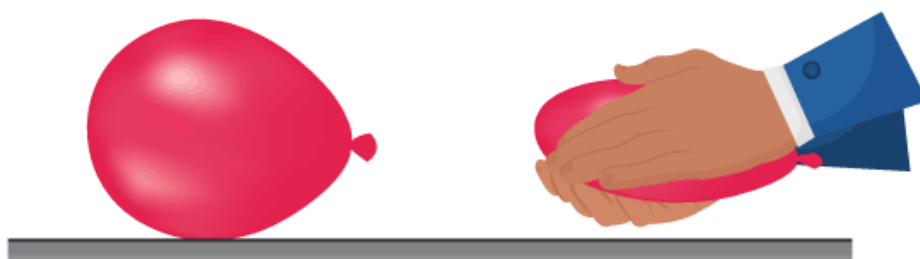
9.1 Introduction to Forces

Definition of Force: A force is a push or pull that can cause an object to move, stop, or change direction.

Forces are everywhere in our daily lives when we open a door, kick a ball, or even when we sit still. Force can make objects move faster, slower, or change shape.

Effects of Force:

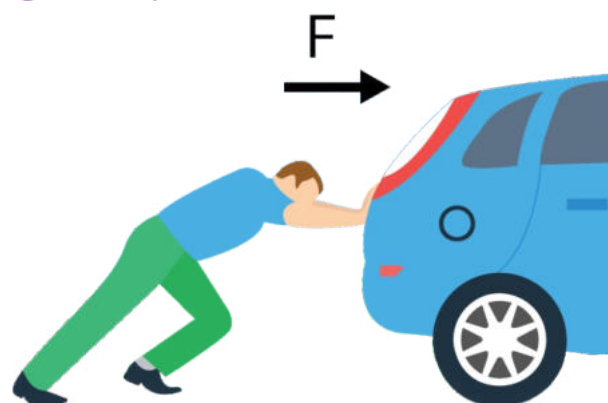
- **Changing shape:** When you press a balloon, it changes shape.
- **Changing size:** When you stretch a rubber band, it becomes longer.
- **Changing motion:** When you push a car, it moves from one place to another.



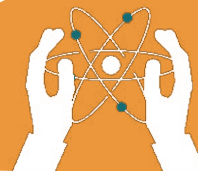
Force that changes shape



Elongate a rubber band



Pushing a car



9.2 Elasticity and Hooke's Law

Have you ever stretched a rubber band or compressed a spring? If you let go, they return to their original shape! This happens because of Hooke's Law, a rule that explains how materials stretch or compress when a force is applied.

Hooke's Law states:

"The force applied to stretch or compress an elastic object is directly proportional to the extension or compression, as long as the material is not stretched too much."

Elastic and Plastic Deformation:

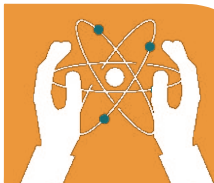
- **Elastic Deformation:** When an object returns to its original shape after the force is removed (like a rubber band).
- **Plastic Deformation:** When an object does not return to its original shape after a force is applied (like a piece of clay).

Springs and Hooke's Law:

- Springs can stretch and return to their original length when the force is removed. This is due to elasticity.

Hooke's Law (Concise Explanation):

Hooke's Law states that the force needed to stretch or compress a spring is directly proportional to the amount of stretch or compression, as long as the spring's elastic limit is not exceeded.



FORCE AND ITS EFFECTS

Formula:

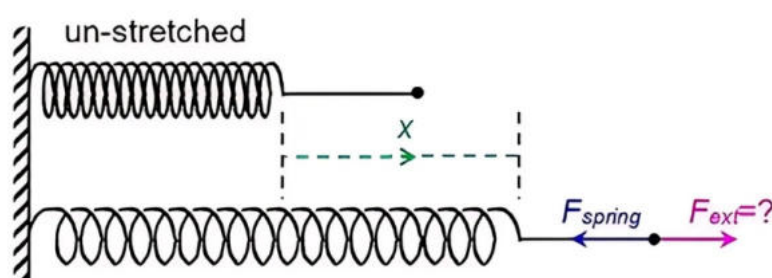
$$F_s = -kx$$

Where:

- **F** = Force applied (in Newtons)
- **k** = Spring constant (stiffness of the spring)
- **x** = Extension or compression (in meters)

Explanation:

The force applied to the spring increases as the spring is stretched or compressed more. A stiffer spring has a higher spring constant.



$$F_s = -kx$$

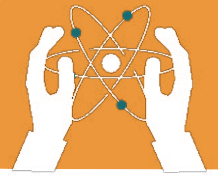
9.3 Pressure

Definition of Pressure: Pressure is the force applied on a surface divided by the area of that surface. Mathematically, it is represented as:

$$P = F/A$$

Where:

- **P** is pressure,
- **F** is force,
- **A** is area.



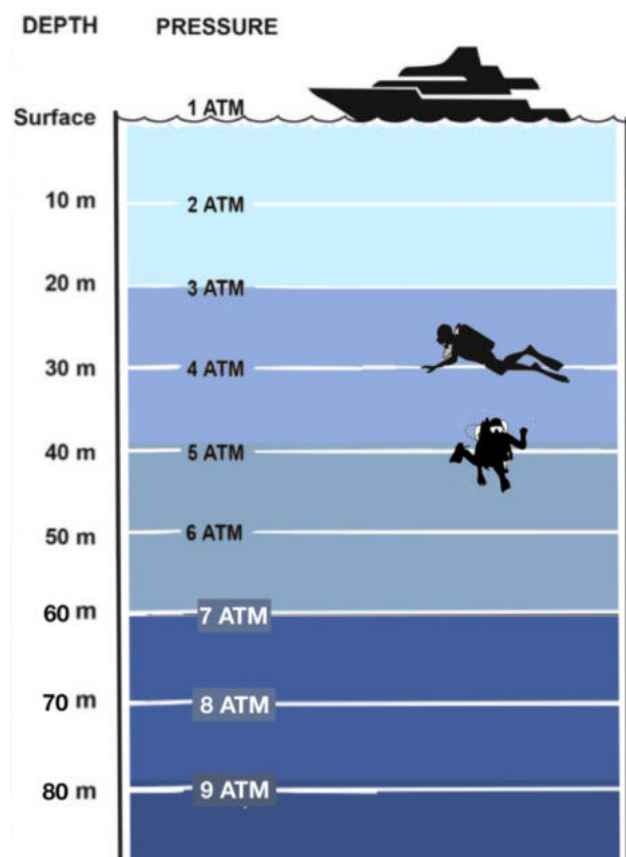
Examples of Pressure in Daily Life:

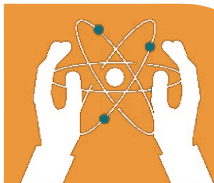
- **High Pressure:** A sharp object like a nail or drawing pin exerts high pressure on a surface because its pointed end has a small area.
- **Low Pressure:** A wide object, like a snowshoe, spreads out its weight over a large area, resulting in lower pressure.

Pressure on Fluids: Pressure increases with depth. For example, when you dive into a pool, the deeper you go, the more pressure you feel because the water above you adds weight.

Factors Affecting Pressure:

- **Depth and Density:** As you go deeper into a fluid, the pressure increases because the density of the fluid and the depth both contribute to the force exerted.





FORCE AND ITS EFFECTS

9.4 Hydraulic Machines and Pascal's Law

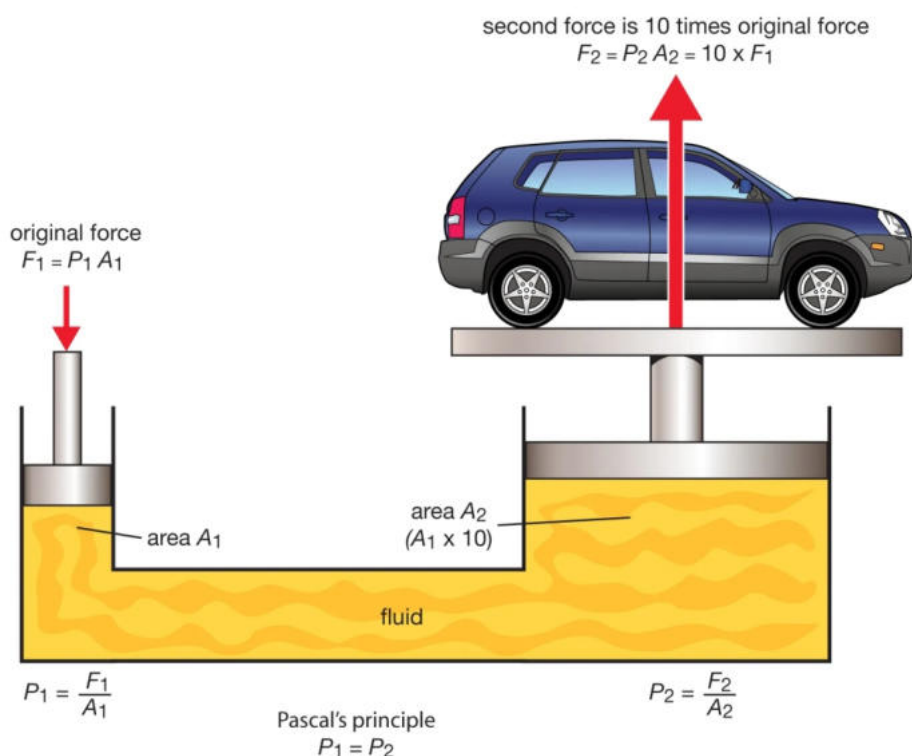
Definition of Hydraulic Machines: Hydraulic machines use liquids (usually oil or water) to transfer force. These machines can lift heavy objects or generate large forces with small amounts of effort.

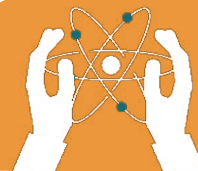
Pascal's Law:

Pascal's Law states that pressure applied to a confined fluid is transmitted equally in all directions. This principle is used in hydraulic systems like car brakes and lifts.

Simple Examples of Hydraulic Systems:

- **Car Brakes:** When you press the brake pedal, the pressure is transmitted through the fluid to the brake pads, stopping the car.
- **Hydraulic Lifts:** When a small force is applied to small piston, the pressure is transferred to a larger piston, lifting heavy objects.

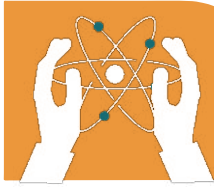




EXERCISES:

Fill in the blanks.

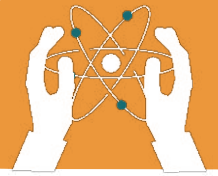
1. A force is a _____ or _____ applied to an object.
2. When a rubber band is stretched and returns to its original shape, it is an example of _____ deformation.
3. The force required to stretch a spring is directly proportional to the _____ the spring is stretched, according to _____ Law.
4. Pressure is calculated by dividing the _____ applied by the _____ area.
5. In hydraulic machines, pressure is transmitted _____ in all directions.
6. A sharp object like a _____ exerts high pressure due to its small surface area.
7. The pressure in a fluid _____ as you go deeper.
8. Pascal's Law applies to _____ systems that use _____ to transmit force.



FORCE AND ITS EFFECTS

Tick(✓) for True and cross (x) for False statements.

1. Force can change the size, shape, and motion of an object. ☐
2. Only solid objects can experience pressure. ☐
3. Elastic deformation occurs when an object does not return to its original shape. ☐
4. A larger area results in higher pressure. ☐
5. Pressure in fluids increases as you go deeper. ☐
6. Hydraulic machines use gas to transfer force. ☐
7. Hooke's Law is applicable only to rubber bands. ☐
8. A wide object will create less pressure than a sharp one. ☐



Choose the correct answer.

1. What is the definition of force?

- a) Push or pull b) Speed or motion
- c) Heat or cold d) Light or darkness

2. Which of the following is an example of elastic deformation?

- a) Clay being shaped
- b) A stretched rubber band returning to its original shape
- c) A piece of wood bending
- d) A glass breaking

3. What happens to pressure as you go deeper into a fluid?

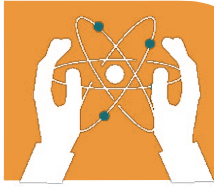
- a) It stays the same b) It decreases
- c) It increases d) It disappears

4. What is used in hydraulic machines to transfer force?

- a) Air b) Water or oil
- c) Electricity d) Heat

5. Which of these is an example of a hydraulic system?

- a) A bicycle pump b) Car brakes
- c) A television remote d) A washing machine



6. What is Pascal's Law about?

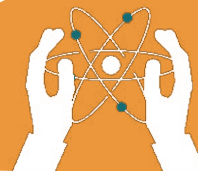
- a) Force is inversely proportional to distance
- b) Pressure is transmitted equally in all directions in a confined fluid
- c) The speed of light is constant
- d) The density of fluids decreases with temperature.

7. What is the unit of pressure?

- a) Newton b) Joule
- c) Pascal d) Meter

8. What type of deformation occurs when an object does not return to its original shape?

- a) Elastic b) Plastic
- c) Reversible d) Permanent



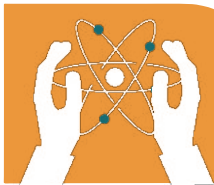
Answer the Following Questions:

Short-Answer Questions.

1. What is force, and how does it affect objects?
2. Write the factors affecting pressure in fluids.

Long-Answer Questions.

1. Explain Hooke's Law and the difference between elastic and plastic deformation.
2. What is pressure? Explain its formula, examples in daily life.
3. What is Pascal's law, and how does it apply to hydraulic lift and car brakes?



FORCE AND ITS EFFECTS



Experiment with Pressure:



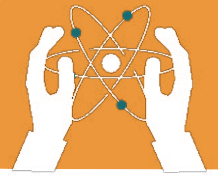
ACTIVITY

Take two objects: a sharp pencil and a flat block. Apply the same force to both objects on a piece of clay or playdough. Measure the depth of indentation and compare the results. Discuss how the area of contact affects the pressure.

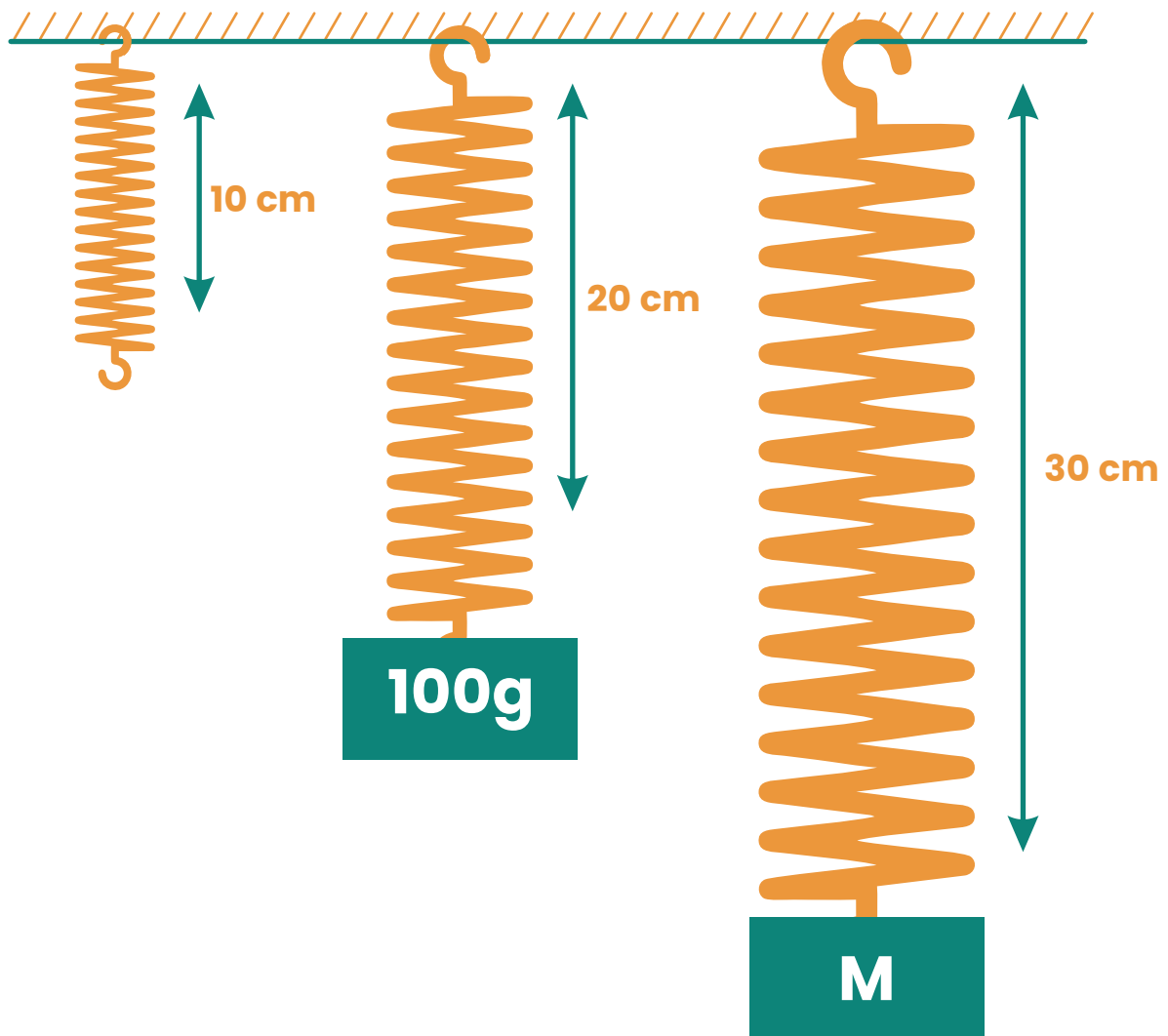


Critical Thinking

1. How does the concept of force and pressure help in designing tools like car brakes or syringes?
2. A truck and a bicycle are moving at the same speed. Suddenly, both apply their brakes at the same time. Which one will stop first? Why?



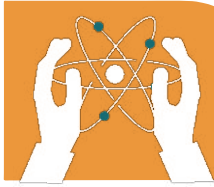
Objects with different masses are hung on a spring. The diagram shows how much the spring stretches.



The extension of the spring is directly proportional to the mass hung on it.

What is the mass of object **M**?

- A. 110g B. 150g C. 200g D. 300g



FORCE AND ITS EFFECTS

Give reasons for the following.

1. Why does a heavier object require more force to move than a lighter object?

2. How does Pascal's Law help in hydraulic brakes?

3. Why do elastic materials return to their original shape after being stretched?
